ALE

1 Conservation Laws on Moving Domain

Let $\Omega_{\boldsymbol{x}}(t) \in \mathbb{R}^{d=3}$ be a time-dependent domain whose spatial point is denoted by $\boldsymbol{x} = (x_1, x_2, x_3)$. In this domain we consider a system of conservation laws written in the Eulerian framework as

$$\left.\frac{\partial \boldsymbol{u}}{\partial t}\right|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{F}(\boldsymbol{u}, \nabla_{\boldsymbol{x}} \boldsymbol{u}) = 0, \quad \text{in } \Omega_{\boldsymbol{x}}(t) \times (0, T], \tag{1}$$

where $\boldsymbol{u} \in \mathbb{R}^m$ is a vector of m conserved variables and $\boldsymbol{F} \in \mathbb{R}^{m \times d}$ contains d flux vectors of dimension m. Examples of conservation laws include the Navier-Stokes equations, the equations of solid motion, and the Maxwell's equations.

2 Transformed Conservation Laws on Reference Domain

2.1 Mapping

Let us denote by $\Omega_{\boldsymbol{X}}$ a fixed (time-independent) reference domain with spatial coordinate $\boldsymbol{X} = (X_1, X_2, X_3)$. Our goal is to transform the system of conservation laws on the moving domain $\Omega_{\boldsymbol{x}}(t)$ to an equivalent system on the reference domain $\Omega_{\boldsymbol{X}}$. This will permit the use of standard discretization techniques to discretize the resulting system on the reference domain. Toward this end we assume that there exists a one-to-one mapping $\boldsymbol{\phi}(\boldsymbol{X}, t)$ that maps every point $\boldsymbol{X} \in \Omega_{\boldsymbol{X}}$ to a point $\boldsymbol{x} \in \Omega_{\boldsymbol{x}}(t)$:

$$\boldsymbol{x} = \boldsymbol{\phi}(\boldsymbol{X}, t). \tag{2}$$

The deformation gradient and velocity are then given by

$$\boldsymbol{G} = \nabla_{\boldsymbol{X}} \boldsymbol{\phi}, \qquad \boldsymbol{v_g} = \left.\frac{\partial \boldsymbol{\phi}}{\partial t}\right|_{\boldsymbol{X}}. \tag{3}$$

Let $g = \det(\boldsymbol{G})$ be the determinant of the deformation gradient $\boldsymbol{G}$. We have that

$$d\Omega_{\boldsymbol{x}} = g d\Omega_{\boldsymbol{X}}, \tag{4}$$

where $d\Omega_{\boldsymbol{x}} = dx_1dx_2dx_3$, $d\Omega_{\boldsymbol{X}} = dX_1dX_2dX_3$ denote volume elements in $\Omega_{\boldsymbol{x}}(t)$ and $\Omega_{\boldsymbol{X}}$, respectively. From this relation, we can obtain the Nanson's formula

$$\boldsymbol{n}_{\boldsymbol{x}} d\mathcal{A}_{\boldsymbol{x}} = g\boldsymbol{G}^{-T}\boldsymbol{n}_{\boldsymbol{X}} d\mathcal{A}_{\boldsymbol{X}}, \tag{5}$$

where $d\mathcal{A}_{\boldsymbol{x}}$ and $d\mathcal{A}_{\boldsymbol{X}}$ denote area elements in $\Omega_{\boldsymbol{x}}(t)$ and $\Omega_{\boldsymbol{X}}$, respectively. Note here that $\boldsymbol{n}_{\boldsymbol{x}}$ and $\boldsymbol{n}_{\boldsymbol{X}}$ are the unit vectors which are outward-pointing and normal to $\mathcal{A}_{\boldsymbol{x}}$ and $\mathcal{A}_{\boldsymbol{X}}$, respectively.

Note that the differentiation of the matrix determinant satisfies the following identity

$$\partial(\det \boldsymbol{A}) = \det \boldsymbol{A}\, \mathrm{tr}(\partial(\boldsymbol{A})\ \boldsymbol{A}^{-1}), \tag{6}$$

where $\mathrm{tr}(\boldsymbol{A}) = A_{ii}$ is the trace of the matrix $\boldsymbol{A}$.

2.2 Piola Transformation

The above mapping has the following property

$$\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{G}^{-T}) = \boldsymbol{0}, \tag{7}$$

which is known as the Piola identity. The proof of the Piola identity is quite simple

$$\int_{\Omega_{\boldsymbol{X}}} \nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{G}^{-T}) d\Omega_{\boldsymbol{X}} = \int_{\partial\Omega_{\boldsymbol{X}}} g\boldsymbol{G}^{-T}\boldsymbol{n}_{\boldsymbol{X}} d\mathcal{A}_{\boldsymbol{X}} = \int_{\partial\Omega_{\boldsymbol{x}}} \boldsymbol{n}_{\boldsymbol{x}} d\mathcal{A}_{\boldsymbol{x}} = \int_{\Omega_{\boldsymbol{x}}} \nabla_{\boldsymbol{x}} \cdot \boldsymbol{I} d\Omega_{\boldsymbol{x}} = \boldsymbol{0}, \tag{8}$$

by the Gauss divergence theorem and the Nanson's formula. For arbitrary tensors $\boldsymbol{W}$ and $\boldsymbol{V}$ we recall the Piola transformation

$$\boldsymbol{W} = g\boldsymbol{V}\boldsymbol{G}^{-T}, \tag{9}$$

and the inverse Piola transformation

$$\boldsymbol{V} = g^{-1}\boldsymbol{W}\boldsymbol{G}^{T}. \tag{10}$$

By the properties of the gradient and divergence operators and using the Piola identity, we obtain

$$\nabla_{\boldsymbol{X}} \cdot \boldsymbol{W} = \nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{V}\boldsymbol{G}^{-T}) = \nabla_{\boldsymbol{X}}\boldsymbol{V} : g\boldsymbol{G}^{-T} + \boldsymbol{V}\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{G}^{-T}) = g(\nabla_{\boldsymbol{X}} \cdot \boldsymbol{V})\boldsymbol{G}^{-T} = g\nabla_{\boldsymbol{x}} \cdot \boldsymbol{V}\ . \tag{11}$$

It thus follows that

$$\nabla_{\boldsymbol{X}} \cdot \boldsymbol{W} = g \nabla_{\boldsymbol{x}} \cdot (g^{-1} \boldsymbol{W} \boldsymbol{G}^T), \qquad \nabla_{\boldsymbol{x}} \cdot \boldsymbol{V} = g^{-1} \nabla_{\boldsymbol{X}} \cdot (g \boldsymbol{V} \boldsymbol{G}^{-T}) \ . \tag{12}$$

We further derive the Piola relationships for arbitrary vectors $\boldsymbol{w}$ and $\boldsymbol{v}$ as follows. Given the Piola tranformation

$$\boldsymbol{w} = g \boldsymbol{G}^{-1} \boldsymbol{v}, \qquad \boldsymbol{v} = g^{-1} \boldsymbol{G} \boldsymbol{w}. \tag{13}$$

we similarly obtain

$$\nabla_{\boldsymbol{X}} \cdot \boldsymbol{w} = \nabla_{\boldsymbol{X}} \cdot (g \boldsymbol{G}^{-1} \boldsymbol{v}) = \nabla_{\boldsymbol{X}} \boldsymbol{v} : g \boldsymbol{G}^{-T} + \boldsymbol{v}^T \nabla_{\boldsymbol{X}} \cdot (g \boldsymbol{G}^{-T}) = g (\nabla_{\boldsymbol{X}} \cdot \boldsymbol{v}) \boldsymbol{G}^{-T} = g \nabla_{\boldsymbol{x}} \cdot \boldsymbol{v} \ . \tag{14}$$

by using the properties of the gradient and divergence operators as well as the Piola identity. It thus follows that

$$\nabla_{\boldsymbol{X}} \cdot \boldsymbol{w} = g \nabla_{\boldsymbol{x}} \cdot (g^{-1} \boldsymbol{G} \boldsymbol{v}), \qquad \nabla_{\boldsymbol{x}} \cdot \boldsymbol{v} = g^{-1} \nabla_{\boldsymbol{X}} \cdot (g \boldsymbol{G}^{-1} \boldsymbol{v}) \ . \tag{15}$$

2.3 Kinematic Relations

We derive here some useful kinematic relations for later use. First, we use the chain rule to obtain the material time derivative

$$\begin{aligned} \left.\frac{\partial \alpha(\boldsymbol{x}, t)}{\partial t}\right|_{\boldsymbol{X}} &= \left.\frac{\partial \alpha}{\partial t}\right|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \alpha \cdot \frac{\partial \boldsymbol{x}}{\partial t} \\ &= \left.\frac{\partial \alpha}{\partial t}\right|_{\boldsymbol{x}} + \boldsymbol{v}_g \cdot \nabla_{\boldsymbol{x}} \alpha, \end{aligned} \tag{16}$$

for any scalar quantity α. For the Jacobian $g = \det(\boldsymbol{G})$, its material time derivative is given by

$$\begin{aligned} \left.\frac{\partial g(\boldsymbol{x}, t)}{\partial t}\right|_{\boldsymbol{X}} &= g \ \mathrm{tr} \left(\left.\frac{\partial \boldsymbol{G}}{\partial t}\right|_{\boldsymbol{X}} \boldsymbol{G}^{-1} \right) \\ &= g \ \mathrm{tr} \left(\left.\frac{\partial}{\partial t}\right|_{\boldsymbol{X}} \frac{\partial \boldsymbol{\phi}}{\partial \boldsymbol{X}} \frac{\partial \boldsymbol{X}}{\partial \boldsymbol{x}} \right) \\ &= g \ \mathrm{tr} \left(\frac{\partial \boldsymbol{v_g}}{\partial \boldsymbol{x}} \right) \\ &= g \ \nabla_{\boldsymbol{x}} \cdot \boldsymbol{v_g} \\ &= \nabla_{\boldsymbol{X}} \cdot (g \boldsymbol{G}^{-1} \boldsymbol{v}_g) \ . \end{aligned} \tag{17}$$

It thus follows that the Jacobian g satisfies

$$\frac{\partial g}{\partial t}\Big|_{\boldsymbol{X}} - \nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{G}^{-1}\boldsymbol{v}_g) = 0 \ . \tag{18}$$

2.4 Transformed Conservation Laws

It follows from the above results to obtain

$$\begin{aligned}
\mathbf{0} &= \frac{\partial \boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{F}(\boldsymbol{u}, \nabla_{\boldsymbol{x}}\boldsymbol{u}) \\
&= \frac{\partial \boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} - \boldsymbol{v}_g \cdot \nabla_{\boldsymbol{x}}\boldsymbol{u} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{F} \\
&= \frac{\partial \boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} - \boldsymbol{v}_g \cdot \nabla_{\boldsymbol{x}}\boldsymbol{u} + g^{-1}\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{F}\boldsymbol{G}^{-T}) \\
&= g^{-1}\frac{\partial g\boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} - g^{-1}\boldsymbol{u}\frac{\partial g}{\partial t}\Big|_{\boldsymbol{X}} - \boldsymbol{v}_g \cdot \nabla_{\boldsymbol{x}}\boldsymbol{u} + g^{-1}\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{F}\boldsymbol{G}^{-T}) \\
&= g^{-1}\frac{\partial g\boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} - \boldsymbol{u}(\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v_g}) - \boldsymbol{v}_g \cdot \nabla_{\boldsymbol{x}}\boldsymbol{u} + g^{-1}\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{F}\boldsymbol{G}^{-T}) \\
&= g^{-1}\frac{\partial g\boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} - \nabla_{\boldsymbol{x}} \cdot (\boldsymbol{u} \otimes \boldsymbol{v_g}) + g^{-1}\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{F}\boldsymbol{G}^{-T}) \\
&= g^{-1}\frac{\partial g\boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} - g^{-1}\nabla_{\boldsymbol{X}} \cdot (g(\boldsymbol{u} \otimes \boldsymbol{v_g})\boldsymbol{G}^{-T}) + g^{-1}\nabla_{\boldsymbol{X}} \cdot (g\boldsymbol{F}\boldsymbol{G}^{-T}) \\
&= g^{-1}\left(\frac{\partial g\boldsymbol{u}}{\partial t}\Big|_{\boldsymbol{X}} + \nabla_{\boldsymbol{X}} \cdot (g(\boldsymbol{F} - \boldsymbol{u} \otimes \boldsymbol{v_g})\boldsymbol{G}^{-T})\right)
\end{aligned} \tag{19}$$

Therefore, we arrive at the transformed system of conservation laws in the reference domain as

$$\frac{\partial (g\boldsymbol{u})}{\partial t}\Big|_{\boldsymbol{X}} + \nabla_{\boldsymbol{X}} \cdot (g(\boldsymbol{F}(\boldsymbol{u}, \nabla_{\boldsymbol{x}}\boldsymbol{u}) - \boldsymbol{u} \otimes \boldsymbol{v_g})\boldsymbol{G}^{-T}) = 0 \ . \tag{20}$$

We now introduce $\boldsymbol{U} = g\boldsymbol{u}$ and thus have

$$\nabla_{\boldsymbol{x}}\boldsymbol{u} = (\nabla_{\boldsymbol{X}}\boldsymbol{u})\boldsymbol{G}^{-1} = (\nabla_{\boldsymbol{X}}(\boldsymbol{U}/g))\boldsymbol{G}^{-1} = (g^{-1}\nabla_{\boldsymbol{X}}\boldsymbol{U} + \boldsymbol{U} \otimes \nabla_{\boldsymbol{X}} g^{-1})\boldsymbol{G}^{-1}. \tag{21}$$

We finally have

$$\frac{\partial \boldsymbol{U}}{\partial t}\Big|_{\boldsymbol{X}} + \nabla_{\boldsymbol{X}} \cdot \boldsymbol{F}_{\boldsymbol{X}}(\boldsymbol{U}, \nabla_{\boldsymbol{X}}\boldsymbol{U}) = 0, \quad \Omega_{\boldsymbol{X}} \times [0, T] \ . \tag{22}$$

where

$$\boldsymbol{F}_{\boldsymbol{X}}(\boldsymbol{U}, \nabla_{\boldsymbol{X}}\boldsymbol{U}) = g(\boldsymbol{F}(g^{-1}\boldsymbol{U}, (g^{-1}\nabla_{\boldsymbol{X}}\boldsymbol{U} + \boldsymbol{U} \otimes \nabla_{\boldsymbol{X}} g^{-1})\boldsymbol{G}^{-1}) - g^{-1}\boldsymbol{U} \otimes \boldsymbol{v_g})\boldsymbol{G}^{-T}. \tag{23}$$

3 Governing Equations for Fluid-Structure Interaction Phenomena

Let $\Omega^f(t)$ and $\Omega(t)$ be two disjoint domains in which particles of fluid and solid are moving as a function of time $t \in [0, t_e]$, respectively. We denote by $\Gamma(t) \equiv \partial\Omega^f \cap \partial\Omega$ the interface on which the interaction between fluid and solid occurs. We also denote by $\Omega(t) \equiv \Omega^f(t) \cup \Omega(t)$ the physical domain which of course varies in time. A point in $\Omega(t)$ will be denoted by $\boldsymbol{x}(t)$. The motion/interaction of fluid and solid are governed by the following systems of equations.

3.1 Equations of Fluid Motion

3.1.1 Compressible Flow

The compressible Navier-Stokes equations in the conservative form are written using the *Eulerian* kinematic balance laws as

$$\left.\frac{\partial \boldsymbol{U}^f}{\partial t}\right|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) = 0, \quad \text{in } \Omega^f(t) \times (0, t_e], \tag{24}$$

where $\boldsymbol{U}^f = (\rho^f, \rho^f \boldsymbol{v}^f, \rho e^f)^T$ is the conservative state vector, ρ^f is the fluid density, $\boldsymbol{v}^f = (u^f, v^f, w^f)^T$ is the fluid velocity field, and e^f is the total internal energy per unit mass.

Moreover, the total flux vectors are given by

$$\boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) = \boldsymbol{A}^{\text{inv}}(\boldsymbol{U}^f) + \boldsymbol{A}^{\text{vis}}(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f), \tag{25}$$

where $\boldsymbol{A}^{\text{inv}}(\boldsymbol{U}^f)$ and $\boldsymbol{A}^{\text{vis}}(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f))$ are inviscid and viscous flux vectors, respectively,

$$\boldsymbol{A}^{\text{inv}}(\boldsymbol{U}^f) = \begin{pmatrix} \rho^f \boldsymbol{v}^f \\ \rho^f \boldsymbol{v}^f \otimes \boldsymbol{v}^f + p^f \mathbf{I} \\ \rho^f (e^f + p^f/\rho^f)\boldsymbol{v}^f \end{pmatrix},$$

$$\boldsymbol{A}^{\text{vis}}(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f)) = -\begin{pmatrix} \mathbf{0} \\ 2\mu^f(\nabla_{\boldsymbol{x}}\boldsymbol{v}^f + (\nabla_{\boldsymbol{x}}\boldsymbol{v}^f)^T) + \lambda^f(\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}^f)\mathbf{I} \\ (\boldsymbol{v}^f)^T\left(2\mu^f(\nabla_{\boldsymbol{x}}\boldsymbol{v}^f + (\nabla_{\boldsymbol{x}}\boldsymbol{v}^f)^T) + \lambda^f(\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}^f)\mathbf{I}\right) + \gamma^f/Pr\nabla_{\boldsymbol{x}}e^f \end{pmatrix}. \tag{26}$$

Here μ^f is the dynamic viscosity coefficient, Pr is the Prandtl number, and $\lambda^f = -\frac{2}{3}\mu^f$ by the Stokes hypothesis. When the fluid obeys the perfect gas equation of state, we have

$$p^f = (\gamma^f - 1)\rho^f(e^f - \frac{1}{2}\boldsymbol{v}^f \cdot \boldsymbol{v}^f), \tag{27}$$

where γ^f is the ratio of specific heats of the fluid.

3.1.2 Sightly Compressible Flow

The slightly compressible Navier-Stokes equations in the conservative form are written using the *Eulerian* kinematic balance laws as

$$\left.\frac{\partial \boldsymbol{U}^f}{\partial t}\right|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}} \boldsymbol{U}^f) = 0, \quad \text{in } \Omega^f(t) \times (0, t_e], \tag{28}$$

where $\boldsymbol{U}^f = (\rho^f, \rho^f \boldsymbol{v}^f, \rho c_p \theta^f)^T$ is the conservative state vector, ρ^f is the fluid density, $\boldsymbol{v}^f = (u^f, v^f, w^f)^T$ is the fluid velocity field, and θ^f is the fluid temperature.

Moreover, the total flux vectors are given by

$$\boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}} \boldsymbol{U}^f) = \boldsymbol{A}^{\text{inv}}(\boldsymbol{U}^f) + \boldsymbol{A}^{\text{vis}}(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}} \boldsymbol{U}^f), \tag{29}$$

where $\boldsymbol{A}^{\text{inv}}(\boldsymbol{U}^f)$ and $\boldsymbol{A}^{\text{vis}}(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}} \boldsymbol{U}^f))$ are inviscid and viscous flux vectors, respectively,

$$\begin{aligned} \boldsymbol{A}^{\text{inv}}(\boldsymbol{U}^f) &= \begin{pmatrix} \rho^f \boldsymbol{v}^f \\ \rho^f \boldsymbol{v}^f \otimes \boldsymbol{v}^f + p^f \mathbf{I} \\ \rho^f (c_p \theta^f + p^f/\rho^f) \boldsymbol{v}^f \end{pmatrix}, \\ \boldsymbol{A}^{\text{vis}}(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}} \boldsymbol{U}^f)) &= - \begin{pmatrix} \mathbf{0} \\ 2\mu^f(\nabla_{\boldsymbol{x}} \boldsymbol{v}^f + (\nabla_{\boldsymbol{x}} \boldsymbol{v}^f)^T) + \lambda^f (\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}^f) \mathbf{I} \\ (\boldsymbol{v}^f)^T \left(2\mu^f(\nabla_{\boldsymbol{x}} \boldsymbol{v}^f + (\nabla_{\boldsymbol{x}} \boldsymbol{v}^f)^T) + \lambda^f (\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}^f) \mathbf{I}\right) + \kappa \nabla_{\boldsymbol{x}} \theta^f \end{pmatrix}. \end{aligned} \tag{30}$$

Here μ^f is the dynamic viscosity coefficient, c_p is the specific heat at constant pressure, κ is the thermal conductivity, and $\lambda^f = -\frac{2}{3}\mu^f$ by the Stokes hypothesis.

We assume that the state equation has the form

$$p^f = g(\rho^f, \theta^f), \tag{31}$$

where g is some given function of the density and temperature. Note that for the special case in which p^f depends only on ρ^f, namely

$$p^f = g(\rho^f), \tag{32}$$

the energy equation is decoupled from the continuity and momentum equations.

3.1.3 Incompressible Flow

The incompressible Navier-Stokes equations in the conservative form are written using the *Eulerian* kinematic balance laws as

$$\begin{aligned} \left.\frac{\partial \boldsymbol{v}^f}{\partial t}\right|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{A}^f(\boldsymbol{v}^f, \nabla_{\boldsymbol{x}} \boldsymbol{v}^f) &= 0, \qquad \text{in } \Omega^f(t) \times (0, t_e], \\ \nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}^f &= 0, \qquad \text{in } \Omega^f(t) \times (0, t_e], \end{aligned} \tag{33}$$

where $\boldsymbol{v}^f = (\boldsymbol{v}^f)^T$ is the conservative state vector and $\boldsymbol{v}^f = (u^f, v^f, w^f)^T$ is the fluid velocity field. In this case, the total flux vectors are given by

$$\boldsymbol{A}^f(\boldsymbol{v}^f, \nabla_{\boldsymbol{x}} \boldsymbol{v}^f) = \boldsymbol{v}^f \otimes \boldsymbol{v}^f + p^f \mathbf{I} - 2\nu^f(\nabla_{\boldsymbol{x}} \boldsymbol{v}^f + (\nabla_{\boldsymbol{x}} \boldsymbol{v}^f)^T), \tag{34}$$

where ν^f is the kinematic viscosity coefficient.

3.2 Equations of Solid Motion

3.2.1 Lagrangian Dynamics

We consider the motion of a solid body $\Omega(t)$ in time $t \in [0, t_e]$. We select the configuration Ω_0 of the body at time $t = 0$, $\Omega_0 = \Omega(t = 0)$, as the material configuration. A point $\boldsymbol{X}$ in Ω_0 shall be used to identify material particles throughout the motion. The motion of the body is described by the deformation mapping

$$\boldsymbol{x}(t) = \boldsymbol{\varphi}(\boldsymbol{X}, t). \tag{35}$$

Thus, $\boldsymbol{x}$ is the position of material particle $\boldsymbol{X}$ at time t.

The Jacobian of the transformation $\boldsymbol{\varphi}$ is the deformation gradient tensor, $\boldsymbol{F}$, and is given by

$$\boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}(\boldsymbol{X}, t)}{\partial \boldsymbol{X}}. \tag{36}$$

The determinant of $\boldsymbol{F}$ is denoted by $J = \det(\boldsymbol{F})$ and relates the volume elements in the material and current configurations. That is, if $dV = dX_1 dX_2 dX_3$ and $dv = dx_1 dx_2 dx_3$, then $dv = J dV$. Since mass is conserved, $\rho dv = \rho_0 dV$, where ρ and ρ_0 are the densities in the current and reference configurations, respectively. Thus, the conservation of mass reduces to $J\rho = \rho_0$.

The material velocity, $\boldsymbol{v}(t)$, and the linear momentum per unit of reference volume, $\boldsymbol{p}$, are given as:

$$\boldsymbol{v} = \frac{\partial \boldsymbol{\varphi}(\boldsymbol{X}, t)}{\partial t}, \qquad \boldsymbol{p} = \rho_0 \boldsymbol{v}. \tag{37}$$

Standard conservation laws in the Lagrangian framework can be readily derived as follows.

3.2.2 Conservation Laws

The conservation of linear momentum is written in the *Lagrangian* framework with respect to the material configuration Ω_0 as

$$\left.\frac{\partial \boldsymbol{p}}{\partial t}\right|_{\boldsymbol{X}} - \nabla_{\boldsymbol{X}} \cdot \boldsymbol{P} = \rho_0 \boldsymbol{b}, \quad \text{in } \Omega_0 \times (0, t_e], \tag{38}$$

where $\boldsymbol{b}(\boldsymbol{X}, t)$ are the body forces per unit mass, and $\boldsymbol{P}(\boldsymbol{X}, t)$ is the first Piola-Kirchhoff stress tensor. The Cauchy stress tensor follows from $\boldsymbol{P}(\boldsymbol{X}, t)$ through the relation

$$\boldsymbol{\sigma} = (J)^{-1} \boldsymbol{P} (\boldsymbol{F})^{-T}, \quad \text{in } \Omega(t) \times (0, t_e]. \tag{39}$$

Conservation of angular momentum requires $\boldsymbol{\sigma}$ to be symmetric.

The conservation of energy is given as follows

$$\left.\frac{\partial e}{\partial t}\right|_{\boldsymbol{X}} + \nabla_{\boldsymbol{X}} \cdot (\boldsymbol{Q} - \frac{1}{\rho_0} (\boldsymbol{P})^T \boldsymbol{p}) = q, \quad \text{in } \Omega_0 \times (0, t_e], \tag{40}$$

where e is the total energy per unit initial volume, $\boldsymbol{Q}$ is the heat flow vector in the material configuration, and q the heat source rate per unit initial volume.

Conservation equation for the deformation gradient $\boldsymbol{F}$ can be easily derived by noting that the time derivative of $\boldsymbol{F}$ is related to the linear momentum vector by

$$\left.\frac{\partial \boldsymbol{F}}{\partial t}\right|_{\boldsymbol{X}} - \frac{1}{\rho_0} \nabla \boldsymbol{p} = 0, \quad \text{in } \Omega_0 \times (0, t_e]. \tag{41}$$

Finally, we recall that conservation of mass is expressed as

$$J \rho = \rho_0, \quad \text{in } \Omega_0 \times (0, t_e]. \tag{42}$$

Later we shall replace this mass conservation law with a new equation that relates J to the so-called hydrodynamic pressure p introduced below. For reversible problems in elastodynamics, the energy equation is decoupled from the momentum equation. In that case, it is sufficient to consider $(\boldsymbol{p}, \boldsymbol{F}, p)$ as problem variables for which we solve.

We would like to point out that the conservation laws for fluid motion are formulated on the time-dependent spatial domain $\Omega^f(t)$, whereas the conservation laws for solid motion are formulated on the time-independent material domain Ω_0. Later we shall unify both the systems of conservation laws on a fixed reference domain $\mathcal{R}$.

3.2.3 Isothermal Hyperelasticity

We first consider the constitutive relation for isothermal hyperelasticity. In this case, the first Piola-Kirchhoff stress tensor is defined through the constitutive equation in terms of the derivative of a strain energy function ψ with respect to the deformation gradient $\boldsymbol{F}$, namely,

$$\boldsymbol{P} = \frac{\partial \psi}{\partial \boldsymbol{F}}. \tag{43}$$

It is customary to use the decoupled representation of the strain energy function ψ in the form

$$\psi = \psi_{\text{iso}}((J)^{-1/3}\boldsymbol{F}) + \psi_{\text{vol}}(J) \qquad \text{with} \qquad \psi_{\text{vol}}(J) = \frac{\kappa}{2}(J-1)^2. \tag{44}$$

This in turn leads to the following relation

$$\boldsymbol{P} = \frac{\partial \psi_{\text{iso}}((J)^{-1/3}\boldsymbol{F})}{\partial \boldsymbol{F}} + pJ(\boldsymbol{F})^{-T}, \tag{45}$$

where

$$p = \kappa(J-1). \tag{46}$$

Once ψ_{iso} is given, we can determine $\boldsymbol{P}$ from (**??**).

As a result, we obtain the following system of equations describing the motion of the structure part

$$\begin{aligned}
\left.\frac{\partial \boldsymbol{F}}{\partial t}\right|_{\boldsymbol{X}} - \frac{1}{\rho_0}\nabla \boldsymbol{p} &= 0, && \text{in } \Omega_0 \times (0, t_e], \\
\left.\frac{\partial \boldsymbol{p}}{\partial t}\right|_{\boldsymbol{X}} - \nabla_{\boldsymbol{X}} \cdot \boldsymbol{P} &= \rho_0 \boldsymbol{b}, && \text{in } \Omega_0 \times (0, t_e], \\
\boldsymbol{P} - \frac{\partial \psi_{\text{iso}}((J)^{-1/3}\boldsymbol{F})}{\partial \boldsymbol{F}} - pJ(\boldsymbol{F})^{-T} &= 0, && \text{in } \Omega_0 \times (0, t_e], \\
\frac{1}{\kappa}p - (J-1) &= 0, && \text{in } \Omega_0 \times (0, t_e].
\end{aligned} \tag{47}$$

In this system, $(\boldsymbol{p}, \boldsymbol{F}, \boldsymbol{P}, p)$ are the field variables for which we solve. Note that the last equation ensures the conservation of mass.

3.2.4 Thermal Hyperelasticity

3.2.5 J2-Viscoplasticity

3.3 Coupled Boundary Conditions

Along solid-wall boundaries, the particle velocity is coupled to the rigid or flexible structure and no fluid particles can cross the fluid-structure interface. Due to the interaction between fluid and structure, boundary conditions on the interface are needed to ensure that the fluid and structural domains will not detach or overlap during the motion. This requirement is satisfied by enforcing the continuity of the linear momentum and of the normal component of the stress along the interface:

$$\begin{array}{rcll} \rho \boldsymbol{v}^f - \boldsymbol{p} & = & 0, & \text{on } \Gamma(t), \\ \boldsymbol{\sigma}^f \cdot \boldsymbol{n}^f + \boldsymbol{\sigma}^s \cdot \boldsymbol{n} & = & 0, & \text{on } \Gamma(t), \end{array} \tag{48}$$

where the Cauchy stress tensor $\boldsymbol{\sigma}^s$ is already defined and the fluid stress $\boldsymbol{\sigma}^f$ is given by

$$\boldsymbol{\sigma}^f = 2\mu^f (\nabla_{\boldsymbol{x}} \boldsymbol{v}^f + (\nabla_{\boldsymbol{x}} \boldsymbol{v}^f)^T) - p\mathbf{I}. \tag{49}$$

Note that the fluid-solid interface $\Gamma(t)$ varies with time. Moreover, since the equations of fluid motion are formulated on the time-dependent domain $\Omega^f(t)$, it proves much more convenient to reformulate the problem on a fixed reference domain $\mathcal{R}$.

4 Reference Domain Formulation of the Fluid-Structure Interaction Problem

We consider an arbitrary *time-independent* reference domain $\mathcal{R}$ which is decomposed into two disjoint parts, $\mathcal{R}^f$ and $\mathcal{R}$, such that $\mathcal{R} = \mathcal{R}^f \cup \mathcal{R}$. Let $\Sigma = \partial\mathcal{R}^f \cap \partial\mathcal{R}$ be the interface between $\mathcal{R}^f$ and $\mathcal{R}$. The discretization of the fluid-structure interaction problem will be carried out in this new reference domain. It is often the case that $\mathcal{R}$ is selected to coincide with the spatial domain $\Omega(t_0)$ at time $t = t_0$ for some $t_0 \in [0, t_e]$.

4.1 Preliminaries

Let $\boldsymbol{n}$, $\boldsymbol{N}$, and $\boldsymbol{\mathcal{N}}$ be the outward unit normals in $\Omega(t)$, $\Omega_0 \equiv \Omega(t=0)$, and $\mathcal{R}$ respectively. We assume that there exists a one-to-one time-dependent mapping $\boldsymbol{\phi}(t)$ between $\mathcal{R}$ and $\Omega(t)$ and that a one-to-one time-independent mapping $\boldsymbol{\psi}$ between $\mathcal{R}$ and Ω_0. Note that if $\mathcal{R} = \Omega_0$ then $\boldsymbol{\psi} = \boldsymbol{I}$ is an identity mapping. Thus a point $\boldsymbol{\mathcal{X}} \in \mathcal{R}$ is mapped to a point $\boldsymbol{x} \in \Omega(t)$ and to a point $\boldsymbol{X} \in \Omega_0$ as

$$\boldsymbol{x} = \boldsymbol{\phi}(\boldsymbol{\mathcal{X}}, t), \qquad \boldsymbol{X} = \boldsymbol{\psi}(\boldsymbol{\mathcal{X}}). \tag{50}$$

In order to transform the Navier-Stokes equations from the fluid physical domain $\Omega^f(t)$ to the fluid reference domain $\mathcal{R}^f$ and the structure equations from the undeformed domain Ω_0 to the structure reference domain $\mathcal{R}$, we require some differential properties of the mapping.

To that end, we introduce the mapping deformation gradients and the mapping velocity as

$$\boldsymbol{G} = \nabla_{\boldsymbol{\mathcal{X}}} \boldsymbol{\phi}, \qquad \boldsymbol{H} = \nabla_{\boldsymbol{\mathcal{X}}} \boldsymbol{\psi}, \qquad \boldsymbol{c} = \left.\frac{\partial \boldsymbol{\phi}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}}. \tag{51}$$

Let $g = \det(\boldsymbol{G})$ and $h = \det(\boldsymbol{H})$ be the determinants of the deformation gradients $\boldsymbol{G}$ and $\boldsymbol{H}$, respectively. We have that

$$dv = g d\mathcal{V}, \qquad dV = h d\mathcal{V}, \tag{52}$$

where $dv = dx_1 dx_2 dx_3$, $dV = dX_1 dX_2 dX_3$, and $d\mathcal{V} = d\mathcal{X}_1 d\mathcal{X}_2 d\mathcal{X}_3$ denote volume elements in $\Omega(t)$, Ω_0, and $\mathcal{R}$, respectively. From this relation, we can obtain an expression for the area change as

$$\boldsymbol{n} da = g \boldsymbol{G}^{-T} \boldsymbol{\mathcal{N}} d\mathcal{A}, \qquad \boldsymbol{N} dA = h \boldsymbol{H}^{-T} \boldsymbol{\mathcal{N}} d\mathcal{A}, \tag{53}$$

where da, dA, and $d\mathcal{A}$ denote area elements in $\Omega(t)$, Ω_0, and $\mathcal{R}$, respectively.

In the sequel, we shall make frequent use of the classical Reynolds transport theorem:

$$\begin{aligned} \frac{d}{dt} \int_{\Omega(t)} \boldsymbol{\alpha} &= \int_{\Omega(t)} \left.\frac{\partial \boldsymbol{\alpha}}{\partial t}\right|_{\boldsymbol{x}} + \int_{\partial\Omega(t)} \boldsymbol{\alpha}(\boldsymbol{c} \cdot \boldsymbol{n}), \\ \frac{d}{dt} \int_{\Omega_0} \boldsymbol{\alpha} &= \int_{\Omega_0} \left.\frac{\partial (\boldsymbol{\alpha})}{\partial t}\right|_{\boldsymbol{X}}, \end{aligned} \tag{54}$$

as well as the generalized Reynolds transport theorem:

$$\begin{aligned} \frac{d}{dt}\int_{\Omega(t)} \boldsymbol{\alpha} &= \int_{\mathcal{R}} \frac{\partial(g\boldsymbol{\alpha})}{\partial t}\Big|_{\boldsymbol{\mathcal{X}}}, \\ \frac{d}{dt}\int_{\Omega_0} \boldsymbol{\alpha} &= \int_{\mathcal{R}} \frac{\partial(h\boldsymbol{\alpha})}{\partial t}\Big|_{\boldsymbol{\mathcal{X}}}. \end{aligned} \tag{55}$$

We are ready to reformulate the governing equations on the reference domain $\mathcal{R}$ as follows.

4.2 Fluid Motion in the Reference Domain

We consider the compressible Navier-Stokes equations written in the spatial domain as

$$\frac{\partial \boldsymbol{U}^f}{\partial t}\Big|_{\boldsymbol{x}} + \nabla_{\boldsymbol{x}} \cdot \boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) = 0, \quad \text{in } \Omega^f(t) \times (0, t_e], \tag{56}$$

In order to obtain the corresponding conservation law written in the reference domain we re-write the above equation in an integral form as

$$\int_{\Omega^f(t)} \frac{\partial \boldsymbol{U}^f}{\partial t}\Big|_{\boldsymbol{x}} + \int_{\partial\Omega^f(t)} \boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) \cdot \boldsymbol{n} = 0, \tag{57}$$

by means of the Gauss divergence theorem.

We next transform the above integrals to the ones on $\mathcal{R}$ as

$$\begin{aligned} \int_{\Omega^f(t)} \frac{\partial \boldsymbol{U}^f}{\partial t}\Big|_{\boldsymbol{x}} &= \frac{d}{dt}\int_{\Omega^f(t)} \boldsymbol{U}^f - \int_{\partial\Omega^f(t)} \boldsymbol{U}^f(\boldsymbol{c}\cdot\boldsymbol{n}) & (58) \\ &= \int_{\mathcal{R}^f} \frac{\partial(g\boldsymbol{U}^f)}{\partial t}\Big|_{\boldsymbol{\mathcal{X}}} - \int_{\partial\mathcal{R}^f} \boldsymbol{U}^f(\boldsymbol{c}\cdot g\boldsymbol{G}^{-T}\boldsymbol{\mathcal{N}}), & (59) \end{aligned}$$

and

$$\int_{\partial\Omega^f(t)} \boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) \cdot \boldsymbol{n} = \int_{\partial\mathcal{R}^f} \boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) \cdot (g\boldsymbol{G}^{-T}\boldsymbol{\mathcal{N}}), \tag{60}$$

where we have used the Reynolds transport theorem and the identity (**??**). As a result, we obtain the following integral equation

$$\int_{\mathcal{R}^f} \frac{\partial(g\boldsymbol{U}^f)}{\partial t}\Big|_{\boldsymbol{\mathcal{X}}} + \int_{\partial\mathcal{R}^f} (\boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) - \boldsymbol{U}^f \otimes \boldsymbol{c}) \cdot (g\boldsymbol{G}^{-T}\boldsymbol{\mathcal{N}}) = 0. \tag{61}$$

Applying the Gauss divergence theorem to this equation we get

$$\frac{\partial(g\boldsymbol{U}^f)}{\partial t}\Big|_{\boldsymbol{\mathcal{X}}} + \nabla_{\boldsymbol{\mathcal{X}}} \cdot g(\boldsymbol{A}^f(\boldsymbol{U}^f, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) - \boldsymbol{U}^f \otimes \boldsymbol{c})\boldsymbol{G}^{-T} = 0, \quad \text{in } \mathcal{R}^f \times (0, T_e]. \tag{62}$$

However, this system is not conservative with respect to the field variable $\boldsymbol{U}^f$.

To write the above equation into a conservative form we introduce a new conservative state vector and flux vectors as

$$\boldsymbol{U} = g\boldsymbol{U}^f, \qquad \boldsymbol{B}(\boldsymbol{U}, \nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{U}) = \boldsymbol{B}^{ins}(\boldsymbol{U}) + \boldsymbol{B}^{vis}(\boldsymbol{U}, \nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{U}). \tag{63}$$

where

$$\begin{aligned} \boldsymbol{B}^{inv}(\boldsymbol{U}) &= g(\boldsymbol{A}^{inv}(\boldsymbol{U}/g, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f) - \boldsymbol{U}/g \otimes \boldsymbol{c})\boldsymbol{G}^{-T}, \\ \boldsymbol{B}^{vis}(\boldsymbol{U}, \nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{U}) &= g(\boldsymbol{A}^{vis}(\boldsymbol{U}/g, \nabla_{\boldsymbol{x}}\boldsymbol{U}^f))\boldsymbol{G}^{-T}, \end{aligned} \tag{64}$$

and

$$\nabla_{\boldsymbol{x}}\boldsymbol{U}^f = (\nabla_{\boldsymbol{\mathcal{X}}}(g^{-1}\boldsymbol{U}))\boldsymbol{G}^{-1} = (g^{-1}\nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{U} + \boldsymbol{U} \otimes \nabla_{\boldsymbol{\mathcal{X}}} g^{-1})\boldsymbol{G}^{-1}. \tag{65}$$

We thus arrive at the following conservative form in the reference domain

$$\left.\frac{\partial \boldsymbol{U}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} + \nabla_{\boldsymbol{\mathcal{X}}} \cdot \boldsymbol{B}(\boldsymbol{U}, \nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{U}) = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e]. \tag{66}$$

It remains to derive a conservative system for the incompressible case.

The momentum equations of the incompressible Navier-System admits the same derivation as shown above. In particular, we introduce $\boldsymbol{v} = g\boldsymbol{v}^f$ and $p = gp^f$ and thus obtain

$$\left.\frac{\partial \boldsymbol{v}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} + \nabla_{\boldsymbol{\mathcal{X}}} \cdot \boldsymbol{B}(\boldsymbol{v}, \nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{v}) = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{67}$$

where

$$\boldsymbol{B}(\boldsymbol{v}, \nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{v}) = g\big(\boldsymbol{v}/g \otimes (\boldsymbol{v}/g - \boldsymbol{c}) + p/g\mathbf{I} - 2\nu^f(\nabla_{\boldsymbol{x}}\boldsymbol{v}^f + (\nabla_{\boldsymbol{x}}\boldsymbol{v}^f)^T)\big)\boldsymbol{G}^{-T}, \tag{68}$$

and

$$\nabla_{\boldsymbol{x}}\boldsymbol{v}^f = (\nabla_{\boldsymbol{\mathcal{X}}}(g^{-1}\boldsymbol{v}))\boldsymbol{G}^{-1} = (g^{-1}\nabla_{\boldsymbol{\mathcal{X}}}\boldsymbol{v} + \boldsymbol{v} \otimes \nabla_{\boldsymbol{\mathcal{X}}} g^{-1})\boldsymbol{G}^{-1}. \tag{69}$$

Moreover, by applying the same transformation to the divergence-free equation $\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}^f$ we get

$$\left.\frac{\partial g}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} + \nabla_{\boldsymbol{\mathcal{X}}} \cdot \boldsymbol{G}^{-1}(\boldsymbol{v} - g\boldsymbol{c}) = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e]. \tag{70}$$

These equations define the conservative system for incompressible fluids in the reference domain.

4.3 Solid Motion in the Reference Domain

When the reference domain $\mathcal{R}$ is selected to coincide with the initial domain Ω_0, the equations for solid motion remain the same as before. Let us consider the general case in which $\mathcal{R}$ is different from Ω_0. Consider the momentum equation

$$\left.\frac{\partial \boldsymbol{p}}{\partial t}\right|_{\boldsymbol{X}} - \nabla_{\boldsymbol{X}} \cdot \boldsymbol{P}(\boldsymbol{F}) = \rho_0 \boldsymbol{b}, \quad \text{in } \Omega_0 \times (0, t_e]. \tag{71}$$

By introducing $\boldsymbol{\varphi} = h\boldsymbol{\varphi}$, $\boldsymbol{F} = \nabla_{\boldsymbol{\mathcal{X}}} \boldsymbol{\varphi}$, $\boldsymbol{p} = h\boldsymbol{p}$, $p = hp$, and $\boldsymbol{b} = h\rho_0\boldsymbol{b}$, we obtain

$$\left.\frac{\partial \boldsymbol{p}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} - \nabla_{\boldsymbol{\mathcal{X}}} \cdot \boldsymbol{P}(\boldsymbol{F}) = \boldsymbol{b}, \quad \text{in } \mathcal{R} \times (0, t_e], \tag{72}$$

where

$$\boldsymbol{P}(\boldsymbol{F}) = h\boldsymbol{P}(\boldsymbol{F})\boldsymbol{H}^{-1}, \tag{73}$$

and

$$\boldsymbol{F} = (\nabla_{\boldsymbol{\mathcal{X}}}(h^{-1}\boldsymbol{\varphi}))\boldsymbol{H}^{-1} = (h^{-1}\boldsymbol{F} + \boldsymbol{\varphi} \otimes \nabla_{\boldsymbol{\mathcal{X}}} h^{-1})\boldsymbol{H}^{-1}. \tag{74}$$

As a result, we obtain the following system of equations

$$\begin{aligned}
\left.\frac{\partial \boldsymbol{F}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} - \frac{1}{\rho_0}\nabla \boldsymbol{p} &= 0, && \text{in } \mathcal{R} \times (0, t_e],\\
\left.\frac{\partial \boldsymbol{p}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} - \nabla_{\boldsymbol{\mathcal{X}}} \cdot \boldsymbol{P} &= \boldsymbol{b}, && \text{in } \mathcal{R} \times (0, t_e],\\
\frac{1}{\kappa}p - h(J-1) &= 0, && \text{in } \mathcal{R} \times (0, t_e],\\
\boldsymbol{p} - \rho_0 \left.\frac{\partial \boldsymbol{\varphi}}{\partial t}\right|_{\boldsymbol{\mathcal{X}}} &= \boldsymbol{0}, && \text{in } \mathcal{R} \times (0, t_e],\\
\boldsymbol{P}(\boldsymbol{F}) - h\boldsymbol{P}(\boldsymbol{F})\boldsymbol{H}^{-1} &= 0, && \text{in } \mathcal{R} \times (0, t_e],\\
\boldsymbol{P} - \frac{\partial \psi_{\text{iso}}((J)^{-1/3}\boldsymbol{F})}{\partial \boldsymbol{F}} - (p/h)J(\boldsymbol{F})^{-T} &= 0, && \text{in } \mathcal{R} \times (0, t_e],\\
\boldsymbol{F} - (h^{-1}\boldsymbol{F} + \boldsymbol{\varphi} \otimes \nabla_{\boldsymbol{\mathcal{X}}} h^{-1})\boldsymbol{H}^{-1} &= 0, && \text{in } \mathcal{R} \times (0, t_e],\\
J - \det \boldsymbol{F} &= 0, && \text{in } \mathcal{R} \times (0, t_e].
\end{aligned} \tag{75}$$

This completes the reference domain formulation.

5 HDG Method

5.1 Problem Statement

The complete set of equations can be written as follows. The equations for fluid velocity are

$$\boldsymbol{L} - \nabla \boldsymbol{v} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{76a}$$

$$\frac{\partial \boldsymbol{v}}{\partial t} + \nabla \cdot \boldsymbol{B} - \boldsymbol{b} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{76b}$$

$$\frac{\partial g}{\partial t} + \nabla \cdot \boldsymbol{G}^{-1}(\boldsymbol{v} - g\boldsymbol{c}) = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e] \tag{76c}$$

$$\boldsymbol{B} - g\big(\boldsymbol{v}/g \otimes (\boldsymbol{v}/g - \boldsymbol{c})\big)\boldsymbol{G}^{-T} - \boldsymbol{P} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{76d}$$

$$\boldsymbol{P} - g\big(p\mathbf{I} - 2\nu(\nabla_{\boldsymbol{x}}\boldsymbol{v} + (\nabla_{\boldsymbol{x}}\boldsymbol{v})^T)\big)\boldsymbol{G}^{-T} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{76e}$$

$$\nabla_{\boldsymbol{x}}\boldsymbol{v} - (g^{-1}\boldsymbol{L} + \boldsymbol{v} \otimes \nabla g^{-1})\boldsymbol{G}^{-1} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e]. \tag{76f}$$

The equations for solid velocity are

$$\frac{\partial \boldsymbol{F}}{\partial t} - \nabla \boldsymbol{v} = 0, \quad \text{in } \mathcal{R}^s \times (0, t_e], \tag{77a}$$

$$\frac{\partial \boldsymbol{v}}{\partial t} + \nabla \cdot \boldsymbol{P} - \boldsymbol{b} = 0, \quad \text{in } \mathcal{R}^s \times (0, t_e], \tag{77b}$$

$$\boldsymbol{P} + \frac{1}{\rho_0}\left(\frac{\partial \psi_{\text{iso}}((J)^{-1/3}\boldsymbol{F})}{\partial \boldsymbol{F}} + pJ(\boldsymbol{F})^{-T}\right) = 0, \quad \text{in } \mathcal{R}^s \times (0, t_e], \tag{77c}$$

$$\frac{1}{\kappa}p - (J - 1) = 0, \quad \text{in } \mathcal{R}^s \times (0, t_e]. \tag{77d}$$

The equations for mesh velocity are

$$\frac{\partial \boldsymbol{F}}{\partial t} - \nabla \boldsymbol{c} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{78a}$$

$$\frac{\partial \boldsymbol{c}}{\partial t} + \nabla \cdot \boldsymbol{P} - \boldsymbol{b} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{78b}$$

$$\boldsymbol{P} + \frac{1}{\rho_0}\left(\frac{\partial \psi_{\text{iso}}((J)^{-1/3}\boldsymbol{F})}{\partial \boldsymbol{F}} + pJ(\boldsymbol{F})^{-T}\right) = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{78c}$$

$$\frac{1}{\kappa}p - (J-1) = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e], \tag{78d}$$

$$\boldsymbol{c} - \frac{\partial \boldsymbol{\phi}}{\partial t} = 0, \quad \text{in } \mathcal{R}^f \times (0, t_e]. \tag{78e}$$

The initial conditions are

$$\boldsymbol{v} = \boldsymbol{v}_0, \qquad \boldsymbol{F} = \boldsymbol{0}, \qquad \boldsymbol{c} = \boldsymbol{c}_0, \qquad \text{on } \mathcal{R} \times \{t = 0\}. \tag{79}$$

The boundary conditions are

$$\begin{aligned} \boldsymbol{v} &= \boldsymbol{g}_D, && \text{on } \partial\mathcal{R}_D \times (0, t_e], \\ \boldsymbol{c} &= \boldsymbol{0}, && \text{on } \partial\mathcal{R}^f \backslash \Sigma \times (0, t_e], \end{aligned} \tag{80}$$

and

$$\boldsymbol{P}\boldsymbol{n} = \boldsymbol{g}_N, \qquad \text{on } \partial\mathcal{R}_N \times (0, t_e]. \tag{81}$$

The coupled boundary conditions are

$$\begin{aligned} \boldsymbol{v}^f - \boldsymbol{v}^s &= \boldsymbol{0}, && \text{on } \Sigma \times (0, t_e], \\ \boldsymbol{c} - \boldsymbol{v}^s &= \boldsymbol{0}, && \text{on } \Sigma \times (0, t_e], \end{aligned} \tag{82}$$

and

$$\boldsymbol{P}^f \boldsymbol{n}^f + \boldsymbol{P}^s \boldsymbol{n}^s = 0, \qquad \text{on } \Sigma \times (0, t_e]. \tag{83}$$

5.2 Approximation Spaces

5.3 Formulation

5.4 Newton-Raphson Procedure

5.5 Solution Strategy